TIC 78186062

Eclipsing binary candidate dossier - Star Vetter

CLEAN

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BLS EPHEMERIS
  Period       4.000878 d
  Epoch (t0)   1543.7815 BTJD
  Duration     0.1700 d
  Depth        9.247 %  (92,470 ppm)
  SNR          98.3

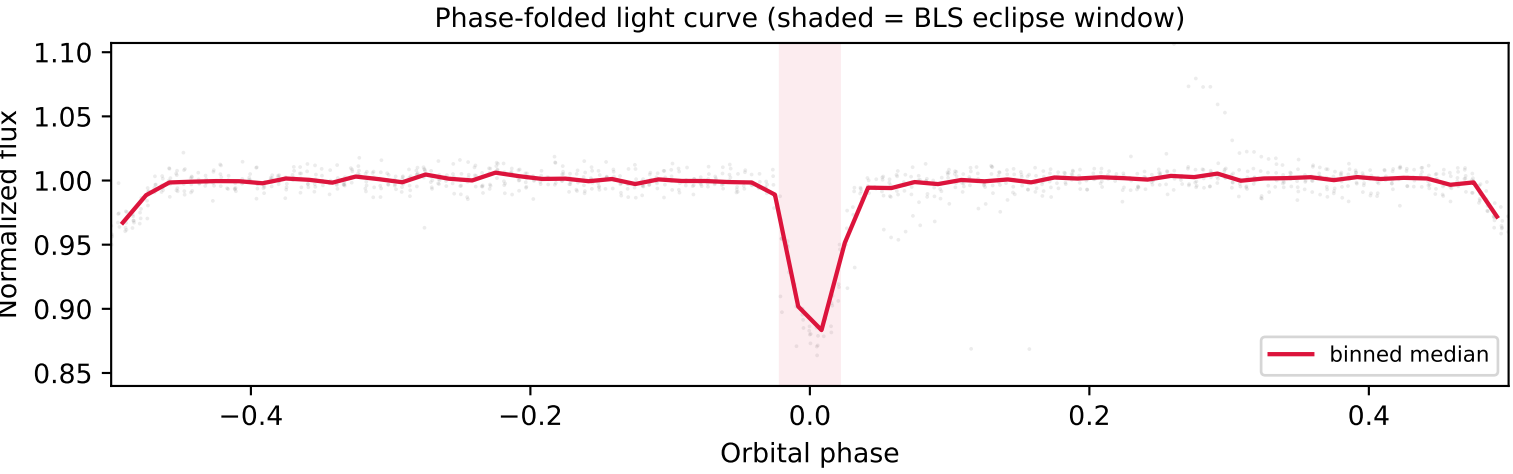
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     5  (gate >= 3)
  Robust depth ratio    1.08  (gate >= 0.50)
  Robust depth sigma    72.7  (gate >= 7.0)
  Odd/even z-score      0.85  (consistent)
  Secondary eclipse     not detected
```

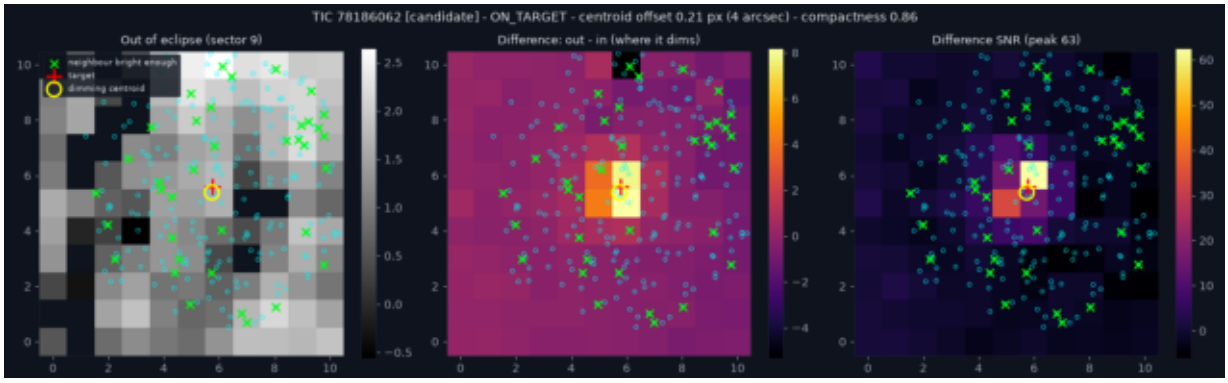
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 9    light-curve dip in this sector: 69.7 sigma    t0 refined by 0.0050 phase
  Difference-image peak SNR   62.60   (gate >= 5.0)      Compactness   0.86   (gate >= 0.50)
  Dimming centroid offset     0.21 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 78186079
    1.05 px away, dTmag +2.19 - farther from the dimming than the target
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No caveats - clean detection on every check.